Miguel G. Silva

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Summary

PhD researcher and applied AI scientist. I work where research and engineering meet: framing the hard modeling problem (e.g., neural surrogates for physics simulation, multi-objective constrained optimization over engineered products), then carrying the answer into production. Strong background in time-series modeling, pattern discovery, and interpretable ML, with first-author publications in Q1 journals.

Work Experience

2025 – Present **Applied AI Scientist** at DareData Engineering

- Co-led the design of a production agentic engineering-to-order loop for a global automotive cable manufacturer, replacing a 16-week / ∼€10k physical prototype-and-lab cycle with a 1-day / ∼€1.8k software workflow (∼80× faster, 5–10× cheaper per RFP).
- Formulated the surrogate-modeling layer: a Graph Neural Network with calibrated uncertainty, trained on multiphysics simulation history, turning hours of physics solves into millisecond inference.
- Designed the multi-objective constrained optimization over a mixed continuous/discrete design space (material × geometry × stack), balancing cost and performance under regulatory and manufacturability constraints. Calls the surrogate as a fast oracle, routing survivors to FEM validation.
- Partnered with the client on product direction: framed engineering requirements as research problems, defined the tools and skills exposed by the engineer assistant, and informed roadmap trade-offs.

2021 – 2026 **Doctoral Researcher** at LASIGE & INESC-ID

- Analyzed smart meter data from 2,170 households using biclustering, uncovering statistically significant and actionable local water consumption patterns inaccessible to traditional clustering methods.
- Developed a scalable methodology to extract population density patterns from mobile phone data (3,743 cells) across Lisbon, uncovering multi-level seasonality and emerging urban trends.
- Designed a method for mining statistically significant motifs in multivariate time series.
- Published 6 first-author papers in top-quartile (Q1) scientific journals.

2022 – 2024 **Teaching Assistant** at IST, University of Lisbon

- Invited TA delivering 140+ hours on Information Retrieval (Python) and Database Systems (SQL) to 300+ students; 4.5/5 average rating across three semesters.

Education

2021 - Expected 2026 **PhD in Computer Science**
Faculty of Sciences University of Lisbon, Portugal

2018 - 2020 **MSc in Data Science**
Faculty of Sciences, University of Lisbon, Portugal
GPA: 18 / 20 , Thesis: 19/20

Technical Skills

Languages & Core: Python · SQL · Git · LaTeX

ML / Deep Learning: PyTorch · Scikit-learn · Optuna · Pandas · NumPy

Applied AI: Surrogate Modeling · Multi-Objective Optimization · Time-Series Modeling · Uncertainty Quantification · Agentic AI

Engineering & MLOps: FastAPI · Pydantic · Docker · CI/CD (GitHub Actions) · MLflow · Async Pipelines

Projects

2026 **Statistical Significance for Multivariate Time Series Motifs** *Publication* | PyPI | GitHub

Open-sourced `msig`, a Python package implementing statistical significance criteria that separate genuine recurring structure from spurious patterns in multivariate time series motif analysis.

2025 **Motif Mining in Multivariate Time Series** *Publication* | GitHub

Built an unsupervised pipeline to mine recurring patterns in residual time series, revealing actionable behavioral insights in energy use, human mobility, and urban activity.

2025 **Next-Motif: Predicting Irregular Pattern Recurrence in Time Series** *Publication* | GitHub

Leveraged LSTM, TCN, and Transformer models to build a forecasting system for irregular event patterns in time series, reducing prediction error by 21–24% on urban mobility and electricity consumption data.

2024 **UrbanFlow: Spatiotemporal Analysis of Population Density Data** *Publication* | GitHub

Developed a Python-based, Docker-deployable tool to extract and visualize spatiotemporal population trends from mobile network data (3,743 cells), enabling actionable insights for data-driven urban planning.

2023 **Biclustering Water Consumption Patterns from Smart Meter Data** Publication

Applied biclustering to household water data from a luxury resort (2,170 users), uncovering high-resolution seasonal consumption profiles to support anomaly detection and data-driven resource planning.